SMART PLANT MONITORING SYSTEM

Arduino-based soil moisture, temperature and humidity monitor

Project Report

Measures soil moisture, ambient temperature and humidity
Displays live readings on an OLED screen
Uses colour status indication and a flashing LED warning

Adapted from the Arduino Project Hub "Smart Plant Soil Monitor" project

Original Arduino project

1. Project Overview

Purpose. The project was built to provide a simple way to monitor the condition of a house plant. An Arduino Uno reads soil moisture, room temperature and air humidity, then presents the results on an OLED display and through visual LED indicators.

Key modification. The original Arduino Project Hub design uses a buzzer for its critical-condition alert. Because a buzzer was not available, I replaced it with a separate LED and changed the software so that the LED flashes every 400 ms when a critical condition is detected. I also implemented timed screen switching so the display cycles through soil moisture, temperature and humidity readings.

2. Project Objectives

- Measure the moisture level of the plant soil and convert the raw sensor value into an easy-to-understand percentage.
- Measure ambient temperature and relative humidity using a DHT11 sensor.
- Display sensor information clearly on a small OLED screen.
- Provide an immediate visual indication of normal, warning and critical conditions.
- Create a visible flashing alert when the plant requires attention.
- Build the system using non-blocking timing so sensing, display updates and alert flashing can operate together.

3. Parts and Software

Component	Purpose
Arduino Uno R3	Main microcontroller and control logic
Analog soil moisture sensor	Measures soil moisture through Arduino analog input A0
DHT11 temperature/humidity sensor	Measures ambient temperature and relative humidity
SSD1306 OLED display (128 x 32)	Displays the three sensor readings
Common-anode RGB LED	Shows normal, warning or critical status
Standard LED	Replaces the original buzzer and flashes during critical conditions
Current-limiting resistors	Protect LEDs from excessive current
Breadboard and jumper wires	Prototype circuit assembly and connections

Component	Purpose
USB cable / 5 V supply	Programming and power
Arduino IDE	Code development, upload and serial monitoring

4. Pin Assignment

The final program uses the following connections:

Device / signal	Arduino connection
Soil moisture sensor	A0
DHT11 data	D2
RGB LED - Red	D3
RGB LED - Green	D4
RGB LED - Blue	D5
Flashing alert LED	D6
OLED SDA	A4 / SDA
OLED SCL	A5 / SCL

5. Build and Development Steps

1. Assemble the prototype circuit: Connected the Arduino, soil sensor, DHT11, OLED and LEDs on a breadboard. The soil sensor was connected to analog input A0, while the temperature/humidity sensor and LEDs used digital pins.

2. Install the required libraries: Added Wire, Adafruit GFX, Adafruit SSD1306 and DHT libraries so the Arduino could communicate with the display and DHT11 sensor.

3. Calibrate the soil sensor: Used dry and wet reference values (900 and 300 in the program) and mapped the raw analog reading to a constrained 0-100% moisture value.

4. Read sensors at a fixed interval: The program takes fresh soil, temperature and humidity readings every 2 seconds and also prints them to the Serial Monitor for testing.

5. Create the status logic: The RGB LED changes colour according to soil moisture and temperature thresholds, allowing the plant condition to be understood at a glance.

6. Replace the buzzer with a flashing LED: Changed the original audible warning to a visual alert. A separate LED toggles every 400 ms whenever a critical condition is active.

7. Cycle the OLED display: The 128 x 32 OLED alternates between soil moisture, temperature and humidity every 10 seconds so each reading remains legible.

8. Test and troubleshoot: Checked sensor readings in the Serial Monitor, verified the OLED output, and tested each LED state by changing moisture conditions and checking temperature thresholds.

6. Software and Control Logic

The soil sensor is calibrated using the following values:

```
const int DRY_VALUE = 900;
const int WET_VALUE = 300;

moisture = map(raw, DRY_VALUE, WET_VALUE, 0, 100);
moisture = constrain(moisture, 0, 100);
```

Status thresholds. The RGB LED provides a quick visual status, while the separate alert LED flashes during a critical condition.

Condition	RGB indication	Alert LED
Moisture >= 40% and temperature 20-30 C	Green	Off
Moisture 30-39%	Blue	Off
Moisture < 30% OR temperature < 20 C OR > 30 C	Red	Flashes every 400 ms

Non-blocking timing. Instead of relying on long delay() calls, the final program uses millis() timers. This allows the sensor read interval, the 10-second screen rotation and the 400 ms flashing alert to run independently.

```
const unsigned long SCREEN_INTERVAL = 10000;
const unsigned long SENSOR_INTERVAL = 2000;
const unsigned long FLASH_INTERVAL = 400;
```

7. Testing and Results

Testing focused on confirming that each part of the system responded correctly. The Serial Monitor was used to compare raw sensor values with the displayed percentage and to verify DHT11 readings.

- Confirmed that soil moisture readings changed when the sensor was moved between drier and wetter soil.
- Confirmed that the OLED cycled between moisture, temperature and humidity readings.
- Confirmed that the RGB LED changed between green, blue and red at the programmed thresholds.
- Confirmed that the replacement alert LED flashed continuously during a critical condition and stopped when the condition cleared.
- Included error handling so the OLED displays “Error” if a DHT11 temperature or humidity reading is unavailable.

Outcome. The completed prototype provides a simple real-time indication of plant and room conditions and alerts the user when the soil is too dry or the temperature falls outside the selected range.

8. Engineering Skills Demonstrated

- Arduino / embedded C++ programming
- Analog and digital sensor integration
- Sensor calibration and data mapping
- I2C communication with an OLED display
- Non-blocking timing with millis()
- Circuit assembly and breadboard prototyping
- Testing, debugging and use of the Serial Monitor
- Adapting an existing design when the specified component was unavailable

9. Possible Future Improvements

- Add an automatic watering pump controlled by a relay or transistor.
- Add Wi-Fi/Bluetooth connectivity and phone notifications.
- Log measurements over time to visualise trends.
- Make moisture thresholds configurable for different plant species.
- Include humidity in the alert decision logic.
- Use a capacitive soil-moisture sensor for improved long-term durability.

10. Reference

Original project: “Smart Plant Soil Monitor”, Arduino Project Hub, published 29 March 2025. The referenced project uses an Arduino Uno, SSD1306 OLED, DHT11, soil moisture sensor, RGB status LEDs and a buzzer. This implementation was adapted by replacing the buzzer with a flashing LED and modifying the display/timing logic. View source project